

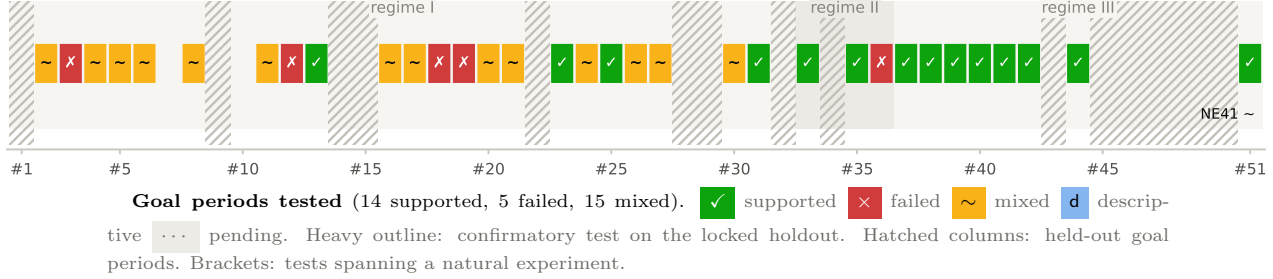
H40: The call clock sets the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 55% Faithfulness 2/5 Usefulness 3/5

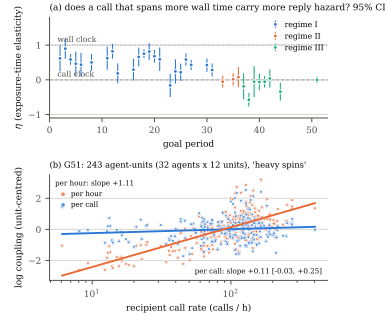
Bottom line. In the computer-use scaffold, agents couple per model call, not per minute: per-hour coupling is a family trait times call rate, and regime I breaks this.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H40 asks whether agents respond to each other per model call or per minute. A reply-hazard model runs on each recipient's real call schedule (context ledger, DQ2 reply labels) in 33 non-holdout periods. Simulated replies first show that the model tells the two clocks apart. In the computer-use scaffold (regimes II–III, 11 periods), coupling runs on the call clock. A call that spans more wall time carries no more reply hazard: $\eta = 0.00 \pm 0.05$ in G51, and -0.03 at timer wakes fixed in advance. In G51 the per-call coupling is flat in cadence ($s = +0.11 [-0.03, 0.25]$) and set by model family. The per-hour coupling scales with call rate (slope 1.0), so slow agents are heavy spins. In regime I (22 periods) the call clock fails: $\eta = +0.51 [0.41, 0.61]$. Natives: G51 yes, NE41 mixed, G18 and NE14 no. Round 2 explains the negative η of G38 and G44 ($-0.57, -0.34$). Long tool calls are followed by fewer talk calls ($\eta_{\text{talk}} = -0.28$). Given a talk call, the span elasticity is $-0.11 [-0.16, -0.06]$ in all four regime-III periods tested. The holdout is not run.



Observables. (a) η per period. Regime-I periods sit between the clocks; regime II–III periods sit at the call clock (0). (b) G51, 243 agent $\times$ unit points. The per-call coupling (blue, model intercepts) is flat in call rate. The per-hour coupling (orange, intercept + $\ln r$) rises with slope ≈ 1 . Lines: fitted slopes.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

In a computer-use agent loop, an agent's wall-time coupling to others is its per-call coupling (a model-family trait) times its call rate. Long tool calls, long pauses and slow models make agents "heavy". In G51 the slowest fifth answer within 5 min $8\times$ less often than the fastest fifth. Cadence is a coupling knob that needs no prompt change. To couple an agent faster, halve its call interval: 5-min coupling rises by about 50% (G51 32%), less at 30 min. Use short pause timers for coordinators. To decouple an agent, give it long batch jobs. The rule does not transfer to a scheduler-driven chat loop (regime I), where time itself carries part of the coupling. The evidence is mostly observational; the clean cadence intervention (NE20) is in the holdout.

PHYSICAL MODEL

Models 02 (kinetic Ising, Glauber attempt rates) and 09. The degree of freedom is the reply of recipient i . Each model call c of i is an update attempt. After i reads message m at call 1, it replies at call n with hazard h_n . α_i is the per-call coupling. e_n is the wall time that call n spans; a_n is the age of m ; b_n counts newer messages (burial); k_n is the batch size. The order parameter η is the exposure-time elasticity: 0 on the call clock, 1 on the wall clock. r_i is the call rate, and $\varepsilon(T)$ the cadence elasticity of a reply within T .

$$\text{cloglog } h_n = \alpha_i + \phi \ln n + \psi \ln a_n + \chi \ln(1+b_n) + \eta \ln e_n + \delta \ln(1+k_n) + \dots$$

$$\text{wall clock: } h_n = 1 - e^{-\lambda e_n} \Rightarrow \eta = 1; \quad \text{call clock: } \eta = 0$$

$$J_i^{\text{hour}} = J_i^{\text{call}} r_i^{1-\eta}, \quad \alpha_i = c + s \ln r_i \quad (s = 0 : \text{per-call invariant})$$

$$\varepsilon(T) = \partial \ln P(\text{reply} \leq T) / \partial \ln(\text{speed-up})$$

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 fra/summaries/build_summaries.py card hypotheses/H40-call-clock-
 coupling/README.md · ratings: H40 agent (suggested 2.5/3.0); coordinator
 calibrated faithfulness to 2.0 (pooled P1 failed, two natives failed); round-2
 update by H40 agent 2026-10-05 (v1 ratings unchanged; v2 left to the coordinator)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 pooled $\eta \in [-0.25, 0.25]$	+0.30 [0.17, 0.43]; II -0.01, III -0.16, I +0.51	fail; II-III yes
P2 decay per call (III)	$\phi = -0.60, \psi = -0.19$; 6/8	yes
P3 $s \approx 0$, excl. -1	pooled -0.12 [-0.28, 0.04]; G51 +0.11	yes
P3 per-hour slope ≥ 0.5	III +0.74; I 0.00	III only
P4 held-out call > wall	33/33	yes
P5 collapse in calls	21/33; test invalid (R5)	uninformative
P6 $\varepsilon(5) \in [0.5, 1]$, $\varepsilon(30)$ lower	0.57 [0.36, 0.78]; 0.47	yes
N1 G51 pause dose, heavy spins	$\eta_1 = -0.03$; R^2 cadence 0.50 vs family 0.33	yes
N2 NE41 no catch-up	read-out $\times 0.44$; ratio 1.86 (band fail)	mixed
N3 G18 / N4 NE14	$\eta = 0.81$ / 1 of 11 agents	no / no
R6 G38/G44: talk composition	$\eta_{\text{talk}} = 0.28$; given talk -0.11 (4 periods); 4 rivals flat	mostly yes
R1 regime-I span split	$\eta_{\text{wait}} = 0.18$ [-0.17, 0.53]; placement +0.04	inconclusive
R5 matched collapse	statistic invalid on synthetic worlds	withdrawn

GOAL PERIODS IN DETAIL

G02 $\sim$ mixed
G03 $\times$ failed
G04 $\sim$ mixed
G05 $\sim$ mixed
G06 $\sim$ mixed
G08 $\sim$ mixed
G11 $\sim$ mixed
G12 $\times$ failed
G13 $\checkmark$ supported
G16 $\sim$ mixed

+24 more periods: see the card.

FAITHFULNESS SCORECARD

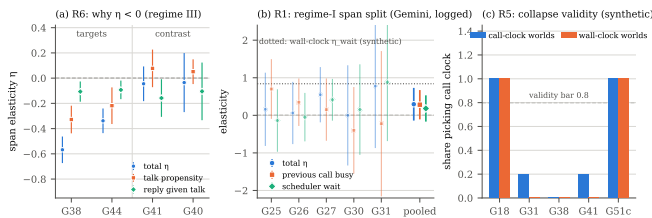
A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Regime I is unexplained. Its positive η survives logged start times, short spans and the placement test. The generation-vs-wait split has only 274 logged replies, too few to decide. The talk split conditions on talk, which a read message can cause (H50). The residual -0.11 given talk has no identified cause. Agents partly choose their cadence; only the pause-timer dose is quasi-exogenous. The cross-sectional G51 slope (1.0) exceeds the model's speed-up elasticity (0.40), so composition adds to cadence. Reply labels are DQ2 parents; rank-2/3 candidates are mostly unlabelled, so multi-target replies are undercounted, most for slow agents with big pools. The NE41 ratio fails for recency reasons found post hoc. The card has two post hoc checks; the holdout is not run.

ROUND 2



Second figure. Round 2. (a) Split of η into talk propensity and reply given talk. In G38 and G44 long calls lower talk; given talk, $\eta \approx -0.1$ everywhere. (b) Regime I, Gemini logged calls: previous-call busy time vs scheduler wait. Pooled η_{wait} stays below the wall-clock value (dotted). (c) Synthetic worlds: the lab-balanced tertile collapse picks the call clock under both clocks, or under neither, so it is withdrawn. Round-1 synthetic: $\hat{\eta}$ recovers 0, 0.5 and 1 (bias ≤ 0.05).

Run `confirm.py`; the NE20 one-tool-call DiD is the cleanest cadence intervention. Test whether the read message causes the talk call before reading -0.11 as the call clock. Repeat the test on the content channel (H29 boundary pull on the ledger).

- **Essence.** a measured coupling-per-call that an operator can multiply by cadence, and an explanation of why the regime-I chat scaffold does not obey it.
- **H40-R1** Why regime I is not a call clock.
- **H40-R2** Content channel.
- **H40-R3** Lags in call cycles (HH187).
- **H40-R4** Run `confirm.py` on NE20/NE44 and the transfer periods; NE20 is the cleanest cadence intervention in the dataset.
- **H40-R5** Fix the collapse test by matching agents on per-call coupling (family) before comparing cadence tertiles, and add multi-parent reply labels (DQ2 holds only the primary parent).
- **H40-R6** Negative η in G38 and G44 (coordinator, 2026-10-04): $\eta = -0.57$ [-0.77, -0.38] and -0.34 [-0.60, -0.08] sit significantly below the call clock, yet the period rule labels them supported.

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